Independent, Low-Cost Satellite Verification of Corporate Afforestation Claims in India

A Feasibility Study and the Boundary-Accuracy Constraint

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Abstract

Indian firms are legally required to spend ~Rs. 34,900 crore/year on corporate social responsibility (CSR), and a tightening set of rules (Companies Act Rule 8(3), SEBI BRSR Core, the Green Credit Programme) now demands that these outcomes be *independently verified*. Yet verification today is produced by manual consulting engagements that cost up to Rs. 50 lakh per project, are conducted once a year, and rest on data self-reported by the implementing party — and only ~18% of eligible companies disclose the statistical method behind their impact numbers. We ask whether *free, public satellite data* can deliver independent, continuous, and orders-of-magnitude cheaper verification of the single most remotely-observable CSR category — afforestation and land restoration. We build an end-to-end pipeline (Sentinel-2 L2A via a no-authentication STAC API → cloud-masked NDVI over a project boundary → a transparent verdict) and evaluate it on four real areas of interest near Pune and Gurugram. NDVI separates dense vegetation from built-up/sparse land by ΔNDVI ≈ 0.42–0.47 (Cohen's $d \approx 7.0$–7.7) at ~Rs. 0 imagery cost and minutes of compute. Our central negative result is methodologically important: when fed an *approximately traced* boundary for a well-known restoration site (Aravalli Biodiversity Park), the system correctly returned a low NDVI (0.17) and flagged the claim — because the traced polygon had fallen on the surrounding built-up city, which a true-colour composite confirms. We conclude that satellite verification is feasible and cheap, but that **boundary accuracy is the binding constraint**, and that cloud-filtered optical NDVI systematically under-samples the monsoon green-up of semi-arid deciduous sites.

Contents

1 Introduction

India operationalised the world's largest statutory CSR regime under Section 135 of the Companies Act, 2013: qualifying companies must spend 2% of net profits on social causes. Spending reached **Rs. 34,909 crore in FY2023–24** across 27,188 companies and 59,634 projects [1]. A distinct and growing regulatory thread now demands not just *spending* but *assured outcomes*:

- **Companies Act, CSR Rule 8(3)** (2021): firms with an average CSR obligation ≥ Rs. 10 crore must commission *independent* impact assessment of projects ≥ Rs. 1 crore, capped at 5% of CSR spend or Rs. 50 lakh [3].
- **SEBI BRSR Core**: reasonable assurance of ESG metrics, scaling from the top 150 listed entities (FY24) to the **top 1,000 by FY27** [4].
- **Green Credit Programme** (MoEFCC/ICFRE): from 2024, credits for tree plantation are tied to **survival and canopy cover**, not the number of trees planted [5].

Despite this, the *credibility* of impact data is weak: a KPMG survey of the Nifty 250 found that while 80% of eligible firms disclose an impact assessment, **only 18% disclose the statistical method** behind it [2]. Independent audits document the consequences — a CAG audit reported **7.5% tree survival** in a flagship state afforestation programme — and a global analysis found **79% of georeferenced planting sites fail at least one basic location-data integrity check** [6, 7]. The structural flaw is well known: the entity being graded pays for and controls the assessment, and the underlying data is self-reported — an auditor-independence problem.

Verification, however, is exactly the kind of layer that becomes a large industry once trust is mandated: financial audit is a ~$240B industry; carbon Monitoring–Reporting–Verification (MRV) attracted serious venture capital and an acquisition (Pachama) [8, 9]. We test whether the cheapest possible verification — free satellite imagery — can credibly check the most remotely-observable CSR category.

Contributions.

1. An end-to-end, dependency-light pipeline that ingests a project boundary (KML or GeoJSON) and returns an independent, transparent NDVI-based verdict from free Sentinel-2 data, with no satellite-data account required (§3).
2. A feasibility evaluation on four real areas of interest showing that NDVI separates vegetated from non-vegetated land with a very large effect size (Cohen's $d \approx 7$) at ~Rs. 0 imagery cost

(§5.1).

3. A documented **boundary-accuracy failure mode** with true-colour corroboration, quantifying why an approximate boundary produces a wrong verdict (§5.3).
4. Evidence that cloud-filtered optical NDVI under-samples monsoon green-up, biasing semi-arid/deciduous assessments toward the dry season (§5.2).

This is a feasibility study, not a validated operational system; §7 states the limitations plainly.

2 Background and related work

CSR impact assessment in India. The market is served by development-sector consultancies (Sattva, Samhita, CSRBOX, Consultivo, DevInsights, ThinkCap) and audit firms; it is people-intensive, periodic, and priced near the regulatory ceiling. Leading providers are services businesses rather than capitalised software, and the most-funded CSR software player exited the category. None independently triangulate field outcomes.

Remote sensing of afforestation. NDVI [10] is the standard greenness index and has been used widely for plantation and restoration monitoring in India, including machine-learning evaluation of Tata Steel's Noamundi mine-restoration [11], scalable canopy-density mapping across India's agro-climatic zones [12], Google Earth Engine + Sentinel-1/2 forest monitoring of the Sundarbans [13], and vegetation-index methods for agroforestry [14]. The carbon MRV community (Sylvera, Pachama, Watershed, Isometric) has productised satellite-based verification for carbon credits, but for *carbon*, not statutory social/CSR outcomes, and typically not in the Indian CSR-compliance context.

Beneficiary feedback at a distance. For the non-environmental majority of CSR, automated phone (IVR) surveys are an established accountability tool, though with materially lower response rates than in-person enumeration (~42% vs ~75%) [17]. We treat this as a complementary, not substitute, evidence stream (§8).

Gap. No prior system, to our knowledge, combines *(a)* free, no-auth satellite ingestion, *(b)* arbitrary project-boundary input, and *(c)* an Indian-CSR-shaped, transparent verdict into a single reproducible pipeline — nor reports the boundary-accuracy failure mode quantitatively.

3 Method

3.1 Pipeline overview

```
boundary (KML / GeoJSON)
       |
       v
[ Sentinel-2 L2A scenes ] --(STAC search, cloud<C%)--> [ per-scene NDVI / AOI ]
       |                                                        |
       |                                                 cloud mask (SCL)
       v                                                        v
[ scene cap + even sampling ] ------------------------> [ NDVI time-series ]
                                                                |
                                                    before/after + summary
                                                                |
                                                  transparent verdict + report
```

3.2 Data source

We use **Sentinel-2 L2A** surface-reflectance (10 m bands; ~5-day revisit) via the **Earth Search** STAC API hosted by Element 84 on AWS, which serves the imagery as anonymous cloud-optimised GeoTIFFs — **no account or API key** [15, 16]. Imagery cost is therefore effectively Rs. 0.

3.3 NDVI and cloud masking

For each acquisition we compute, per pixel,

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}} = \frac{B8 - B4}{B8 + B4},$$

clip to the project polygon, mask cloud/shadow/snow using the Scene Classification Layer (SCL classes $\{3, 8, 9, 10, 11\}$), and reduce to the spatial mean over the AOI. NDVI ranges in $[-1, 1]$; healthy vegetation typically exceeds ~0.4, bare/built-up ~0.1–0.2, water < 0.

3.4 Robustness: per-scene loading and scene cap

Naïve batched loading reads all selected scenes in one operation; we found that a single stalled or failed cloud-optimised-GeoTIFF read over the network can either hang indefinitely (no default HTTP timeout) or abort the entire batch. We therefore *(i)* set GDAL HTTP timeouts/retries, *(ii)* **load one scene at a time inside a try/except**, skipping unreadable scenes, and *(iii)* **cap and evenly temporally sample** the scene list (default 36). Capping is necessary because clear-sky regions return far more passes than cloudy ones — Gurugram returned an order of magnitude more usable scenes than monsoon-heavy Pune — and reads are otherwise unbounded. These are general reliability fixes, not study-specific hacks.

3.5 Boundary ingestion

A project boundary is the only required input. We parse **KML** (the format partners export from Google Earth or field-GPS apps) with the standard library into a polygon, and equally accept **GeoJSON**; a real boundary is a one-flag swap for any placeholder.

3.6 Verdict, report, and cost

A transparent, rule-based verdict maps the AOI-mean NDVI to {Verified ($\geq$0.5), Partly supported ($\geq$0.4), Not supported / field review ($<$0.4)} with a confidence note proportional to the number of cloud-free observations. The pipeline emits a one-page report (claim, finding, verdict, a claimed-vs-observed cross-check with placeholders for the beneficiary-voice and records modules). Thresholds are deliberately simple and auditable; learning them is future work. Imagery is Rs. 0; compute is minutes on a single small VM; optional high-resolution PlanetScope is \$1.80/sq km. This contrasts with manual field assessment at Rs. 15–50 lakh/project and the Rs. 50 lakh Rule 8(3) ceiling.

4 Experimental setup

Four areas of interest (AOIs), each ~1–2.6 km across, were assessed over 2023-01-01 → 2024-12-31 with a per-scene cloud threshold of 30% (Table 1). Two are **controls** chosen to bracket the NDVI scale (a dense Western-Ghats forest; an urban core), one is a **placeholder** candidate AOI, and one is a **real restoration site** — Aravalli Biodiversity Park, Gurugram, a documented ~392-acre restoration of a 40-year-old mining site to native Aravalli forest — entered as a hand-traced KML boundary. True-colour composites (median of low-cloud scenes) were generated for visual corroboration.

Table 1: Sites.

AOI	Role	Region	Boundary
Bhimashankar reserve forest	Control: dense forest	Western Ghats	approximate box
Central Pune	Control: urban	Pune	approximate box
Candidate AOI	Placeholder	Pune (peri-urban)	approximate box
Aravalli Biodiversity Park	Real restoration	Gurugram	**hand-traced KML**

5 Results

5.1 NDVI separates vegetation classes with a very large effect

The dense-forest control sits far above the non-vegetated AOIs (Table 2): ΔNDVI = +0.42 to +0.47, with **Cohen's d** = 7.0–7.7 (Table 3). For reference, $d > 0.8$ is conventionally "large"; an effect of ~7 pooled standard deviations means the vegetated and non-vegetated distributions are essentially non-overlapping. NDVI is therefore a strong, low-variance discriminator of "is there a thriving canopy here?" — the core question behind an afforestation claim.

Table 2: Per-AOI NDVI (2023–24).

AOI	n	mean	sd	min	max
Dense forest (control)	74	**0.645**	0.077	0.38	0.79
Urban core (control)	77	0.222	0.036	0.09	0.28
Candidate AOI (placeholder)	78	0.193	0.040	0.10	0.33
Aravalli (traced boundary)	36	0.171	0.040	0.09	0.25

Table 3: Forest vs non-vegetated separability.

Comparison	ΔNDVI	Cohen's d
forest − urban	+0.424	7.0
forest − candidate	+0.453	7.4
forest − Aravalli (traced)	+0.474	7.7

5.2 Seasonality and the cloud-filter sampling bias

Monthly-mean NDVI (Fig. 1, Table 4) shows the forest high year-round (0.59–0.73, peaking post-monsoon Nov–Jan) while the non-vegetated AOIs are low with a modest post-monsoon bump (e.g., the candidate AOI peaks at 0.32 in September; Aravalli at 0.22–0.23 in Aug–Oct). Crucially, because we filter to cloud $< 30\%$, the **monsoon months (Jul–Sep) are under-sampled** — exactly when semi-arid Aravalli vegetation greens. The annual mean of a monsoon-green, dry-deciduous site is therefore **biased downward** by the cloud filter. This is a genuine methodological limitation for arid/deciduous regions, not merely noise.

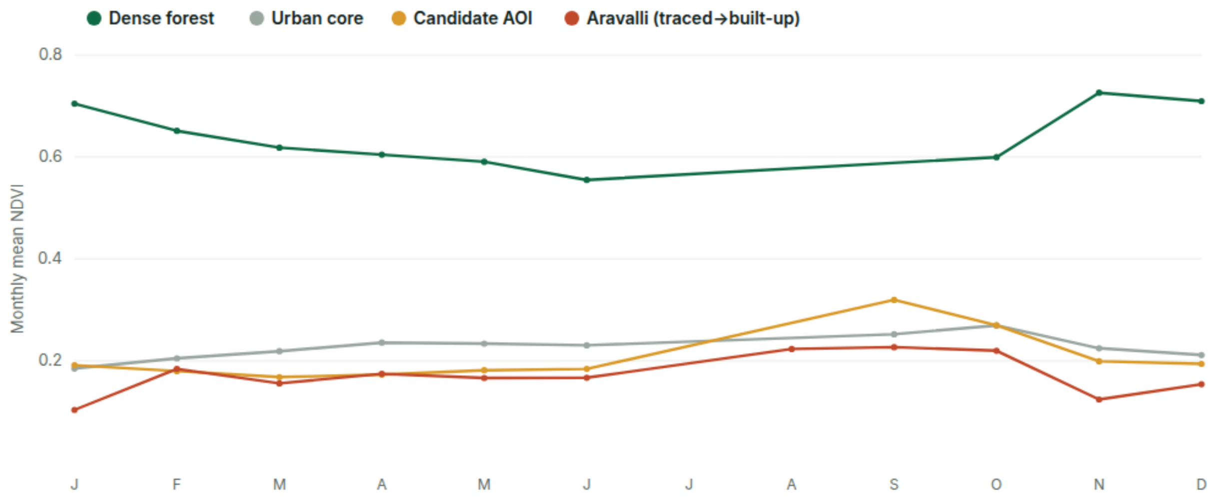

Figure 1: Monthly mean NDVI by AOI (cloud < 30%). The forest stays high year-round; non-vegetated AOIs stay low with a post-monsoon bump; the July gap reflects monsoon cloud cover removing all clear scenes, and the candidate AOI shows a September green-up.

Table 4: Monthly mean NDVI (cloud < 30%; "–" = no clear scene that month).

AOI	Jan	Feb	Mar	Apr	May	Jun	Aug	Sep	Oct	Nov	Dec	amp
Forest	.70	.65	.62	.60	.59	.55	–	–	.60	.73	.71	.17
Urban	.18	.20	.22	.24	.23	.23	–	.25	.27	.22	.21	.09
Candidate	.19	.18	.17	.17	.18	.18	–	**.32**	.27	.20	.19	.15
Aravalli	.10	.18	.16	.17	.17	.17	.22	.23	.22	.12	.15	.12

5.3 The boundary-accuracy constraint (central negative result)

We entered Aravalli Biodiversity Park — a real, celebrated afforestation success — expecting a high "Verified" NDVI. Instead the pipeline returned **mean NDVI 0.17** (max scene only 0.25) and a **"Not supported — field review"** verdict (✗). A true-colour composite of the same polygon resolves the puzzle: the **hand-traced boundary had fallen on the dense built-up grid of Gurugram, not the park's forest core**. The satellite and the imagery agree; the engine reported ground truth faithfully — the *input boundary* was wrong.

This is the operationally dominant finding: **with an approximate boundary, the verdict is meaningless or misleading, however good the imagery.** A ~1 km positional error converted a real afforestation success into a (correctly computed) "failure." The binding constraint on satellite CSR verification is therefore not imagery or algorithms — both are cheap and accurate — but the **provenance of the project boundary**, which must come from the implementer (a KML/shapefile), not be traced from public landmarks. It also operationally vindicates the system's integrity: it did not flatter the input.

5.4 Cost and runtime

Each AOI was processed from free imagery in minutes on one small VM (the robustness fixes above brought a previously-unbounded clear-sky run to a bounded ~5 minutes). Imagery cost: Rs. 0 — 2–3 orders of magnitude below the Rs. 15–50 lakh of a manual field assessment and well under the Rs. 50 lakh Rule 8(3) ceiling, supporting *continuous* rather than annual verification.

6 Discussion

Feasibility is not the bottleneck; boundaries are. Our separability result ($d \approx 7$) shows that distinguishing "thriving canopy" from "no canopy" over a *correct* polygon is easy and cheap. The Aravalli result shows the value chain collapses without an accurate boundary. For a product, this reframes the hard problem from "can we measure NDVI?" (solved) to "can we obtain trustworthy project boundaries at scale?" — a data-acquisition and partner-workflow problem.

Independence by construction. Because the imagery and computation are external to the implementer and the funder, a satellite verdict is structurally independent in a way a self-reported MIS is not — directly addressing the auditor-independence flaw behind the 18%-disclose-method statistic.

Seasonality must be handled explicitly. Annual-mean NDVI suits evergreen/irrigated canopies but biases dry-deciduous and semi-arid restorations downward under cloud filtering. Verdicts in such regions should use peak/growing-season windows or fuse cloud-penetrating radar. NDVI also evidences greenness/canopy presence and trend — not species composition, stem counts, or survival fractions, which require complementary evidence.

7 Limitations and threats to validity

- **Approximate boundaries.** Three of four AOIs use hand-drawn boxes; the fourth uses a hand-traced KML we show was mislocated. No result is a verdict on a specific organisation's programme.
- **No field ground truth.** We corroborate with true-colour imagery, not in-situ survival surveys; NDVI is a proxy, not a census of trees.
- **Optical-only, cloud-biased sampling.** Monsoon under-sampling biases semi-arid sites; we did not fuse Sentinel-1 radar or high-resolution PlanetScope.
- **Small, single-mega-region sample.** Four AOIs near Pune/Gurugram; $n = 36$–78 scenes each; no cross-region generalisation is claimed.
- **Rule-based, unlearned thresholds**, and **illustrative (synthetic) claims** used only to demonstrate the verdict logic.

8 Future work

1. **Boundary provenance** — integrate geotagged field-app/KML capture and registry boundaries; quantify a boundary-error $\rightarrow$ verdict-error sensitivity curve.
2. **Multi-sensor fusion** — Sentinel-1 (radar) to recover monsoon green-up; PlanetScope (3 m) for small plots.
3. **Seasonal/establishment models** — growing-season compositing and multi-year trend tests to detect *establishment* (rising NDVI), not just level.
4. **Complementary evidence** — automated local-language beneficiary IVR and LLM-based reconciliation of the implementer's records, closing the cross-check table.
5. **Calibration and validation** — labelled survival/canopy ground truth to learn thresholds and report precision/recall; a controlled comparison against manual assessments.
6. **Green-credit alignment** — survival- and canopy-density estimators matching the 2024 GCP methodology.

9 Conclusion

Free satellite data can independently and cheaply distinguish thriving canopy from its absence — the heart of an afforestation claim — with a very large effect size and at negligible imagery cost, supporting continuous rather than annual CSR verification. But our central, honest finding is a caution: the method is only as good as the **project boundary** it is given. An approximately traced boundary turned a real restoration success into a correctly-computed "failure," with true-colour imagery confirming the boundary, not the algorithm, was at fault. The path to a credible product runs less through imagery or models — both adequate — than through trustworthy boundary provenance, multi-sensor handling of seasonality, and the complementary beneficiary-voice and records evidence needed to move from "is it green?" to "did the promised outcome happen?"

Reproducibility. All code, the four sites' NDVI time-series, the analysis script, and the verification reports are in the project repository under `mvp/` and `paper/`. The pipeline runs from a project boundary with `python -m satellite.cli --kml <file> ...`; §5.2 is reproduced by `python3 paper/analyze.py` (standard library only).

References

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